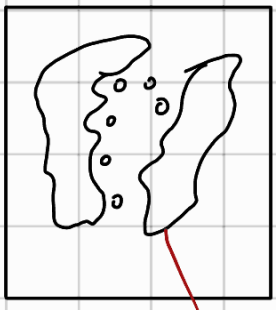
Conditional dilation

$A (+)_C B = (A \oplus B) \cap C$

(c)



opening

$A = C \circ D$ ← D büyük olsun



(dilated)

büyük küp kesişme yapınca C ile

boşalar kalır

Connected component bulmak için

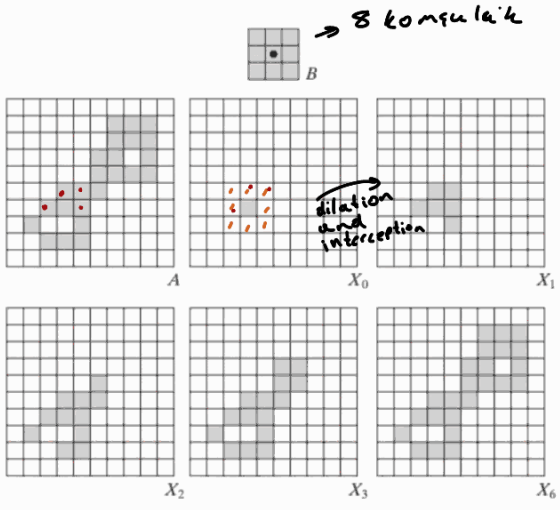
$X_k = (X_{k-1} \oplus B) \cap A$

$k = 1, 2, \ldots$

X_0 ihtiyaçsız, objenin üzerindeki herhangi bir nokta

$$X_k = (X_{k-1} \oplus B) \cap A \qquad k = 1, 2, 3, \ldots \qquad (9.5\text{-}3)$$

where B is a suitable structuring element (as in Fig. 9.17). The procedure terminates when $X_k = X_{k-1}$, with X_k containing all the connected components



a
b c d
e f g

FIGURE 9.17 Extracting connected components. (a) Structuring element. (b) Array containing a set with one connected component. (c) Initial array containing a 1 in the region of the connected component. (d)–(g) Various steps in the iteration of Eq. (9.5-3).

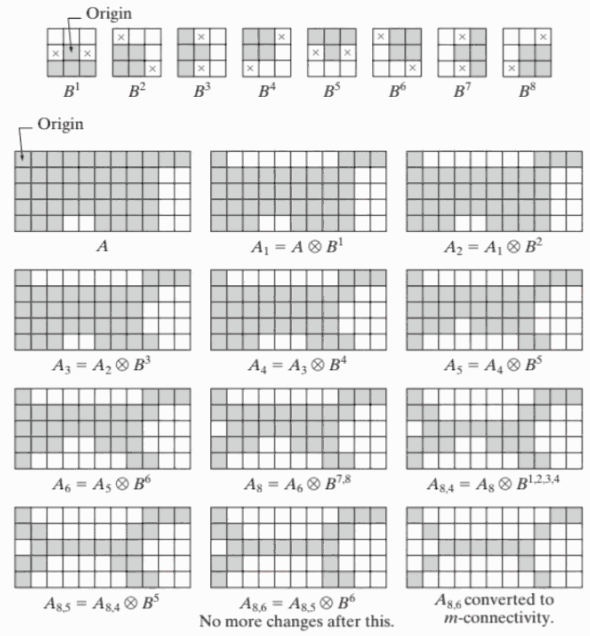
X_0 bulmak için etiketlemeyi

herhangi bir nokta

9.5.6 Thickening

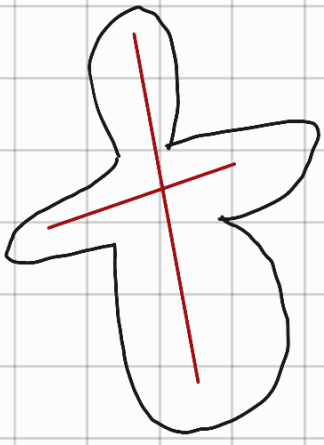
Thickening is the morphological dual of thinning and is defined by the expression

$$A \odot B = A \cup (A \circledast B) \tag{9.5-9}$$

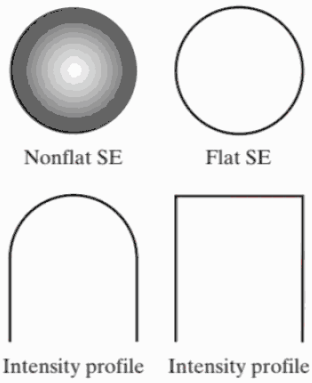


a b c d e f g h i j k l m

FIGURE 9.21 (a) Sequence of rotated structuring elements used for thinning. (b) Set A. (c) Result of thinning with the first element. (d)–(i) Results of thinning with the next seven elements (there was no change between the seventh and eighth elements). (j) Result of using the first four elements again. (l) Result after convergence. (m) Conversion to m-connectivity.



Gray Scale morphology



a b c d

FIGURE 9.34 Nonflat and flat structuring elements, and corresponding horizontal intensity profiles through their center. All examples in this section are based on flat SEs.

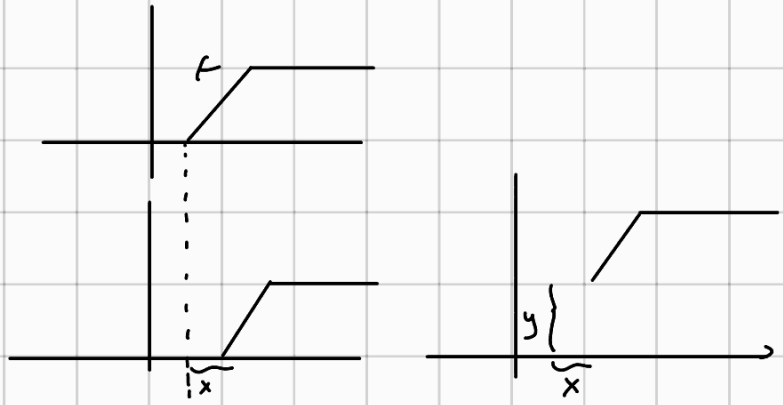
x bir değişken

$f(x)$

$f_x(z) = f(z-x)$ translation by x

$(f+y)(z) = f(z)+y$ offset by y

$f_x + y$



$D[g]$ $D[f]$

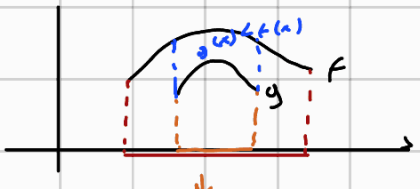
altında

g and f are signals with domain $D[g]$ and $D[f]$ we say that g is beneath f, if

1) $D[g] \subseteq D[f]$

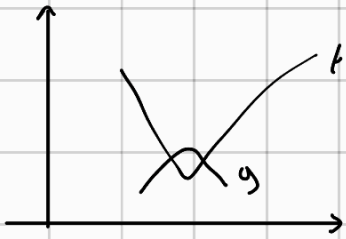
2) $g(x) \leq f(x)$ for any $x \in D[g]$

Then we write $g \ll f$

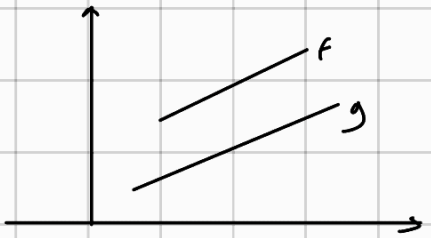


Bu x değerlerinde g belirli sonlu bir değer aldığı bölge $D[g]$

$\rightarrow g \ll f$

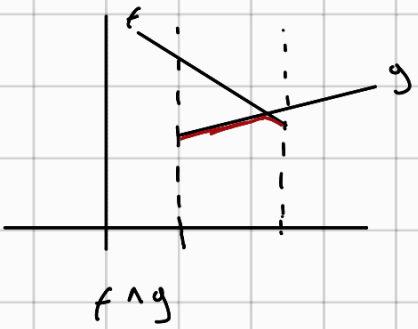


altında değil

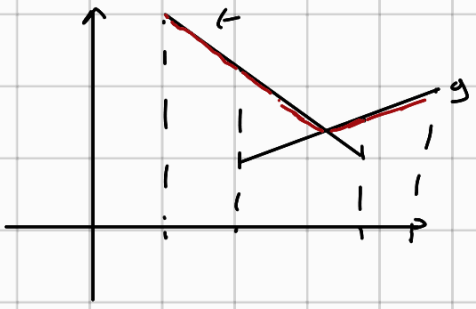


$$(f \wedge g)(x) = \min\{f(x), g(x)\} \quad \text{for } x \in D[f] \cap D[g]$$

$$(f \vee g)(x) = \begin{cases} \max\{f(x), g(x)\} & \text{for } x \in D[g] \cap D[f] \\ f(x) & \text{for } x \in D[f] \text{ and } x \notin D[g] \\ g(x) & \text{for } x \in D[g] \text{ and } x \notin D[f] \\ \text{not defined} & \text{for } x \notin D[g] \text{ and } x \notin D[f] \end{cases}$$

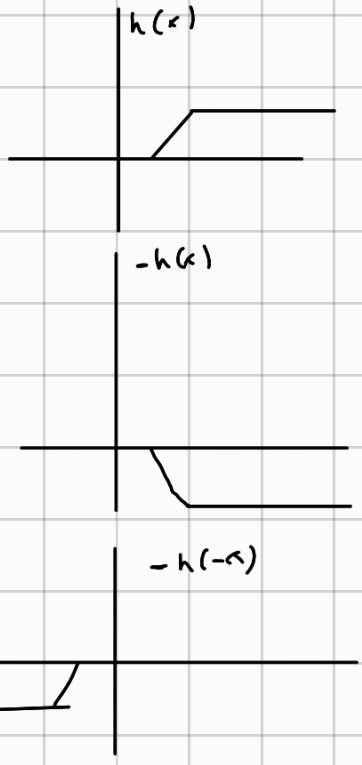


$f \wedge g$



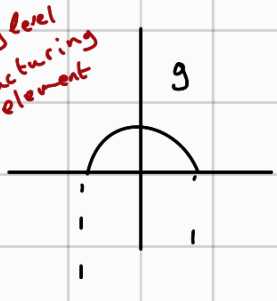
Reflection of h

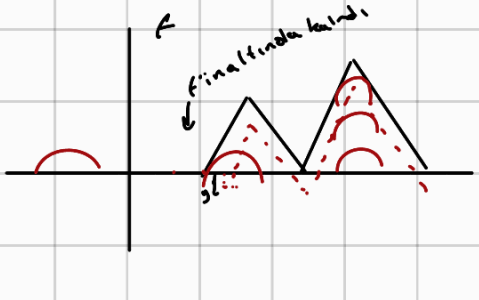
$$\hat{h}(x) = -h(-x)$$

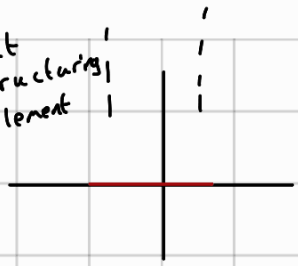


$$(f \ominus g)(x) = \max\{y : g_x + y \leq f\}$$

gray level structuring element

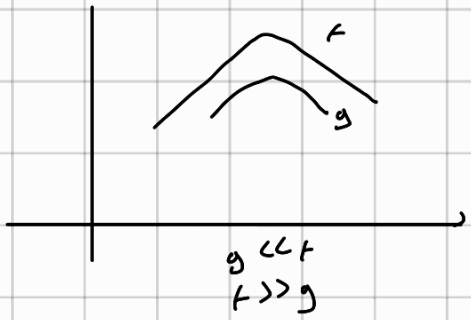






$$(f \oplus g)(x) = \min\{ y : (g^{\vee})_x + y >> f \} = \bigvee^{\max} \{ f_x + g(x) : x \in D[g] \}$$





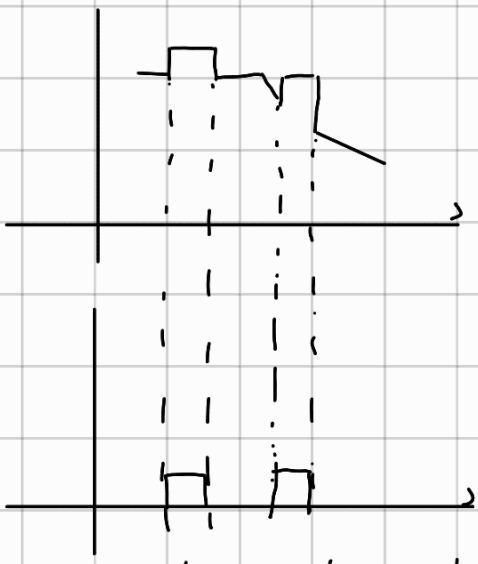
$$f \circ g = (f \ominus g) \oplus g$$

$$f \bullet g = (f \oplus g) \ominus g$$

$$(f \circ g) \circ g = f \circ g$$

$$\text{Grad}(f) = f \oplus g - f \ominus g$$

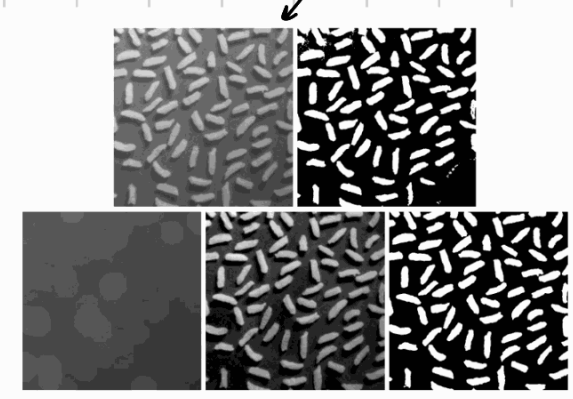
$$\text{Top-Hat}(f) = f - f \circ g$$



opening olayı background estimation olarak yarar

$f \circ g$ background

arkası karanlık üstü beyaz şeyler için



a b
c d e

FIGURE 9.40 Using the top-hat transformation for *shading correction*. (a) Original image of size 600 × 600 pixels. (b) Thresholded image. (c) Image opened using a disk SE of radius 40. (d) Top-hat transformation (the image minus its opening). (e) Thresholded top-hat image.

Bottom Hat (f) = $f - f \circ g$

↓ Gradient

$A \oplus B - A \ominus B$

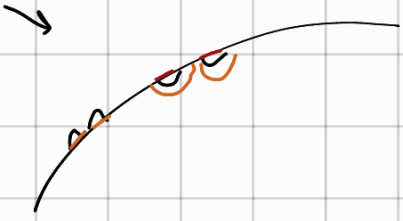
Morphological laplacian $\frac{1}{2}(A \oplus B + A \ominus B - 2A)$

$\frac{1}{2}((A \oplus B - A) - (A - A \ominus B))$

türev türev

↓ smoothing

$(A \circ B) \bullet B$



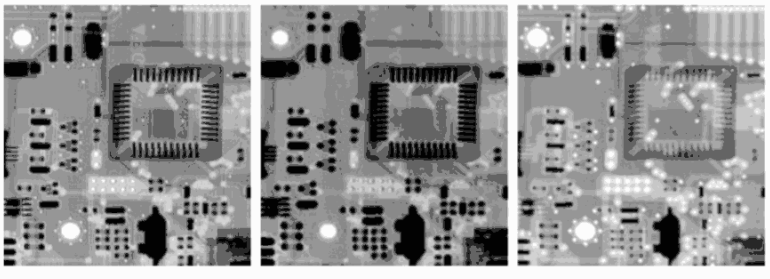
minimum

$\frac{A - A \oplus B}{A \oplus B - A \ominus B}$ } normalizasyon gibi

→ B biraz büyük seçilmeli

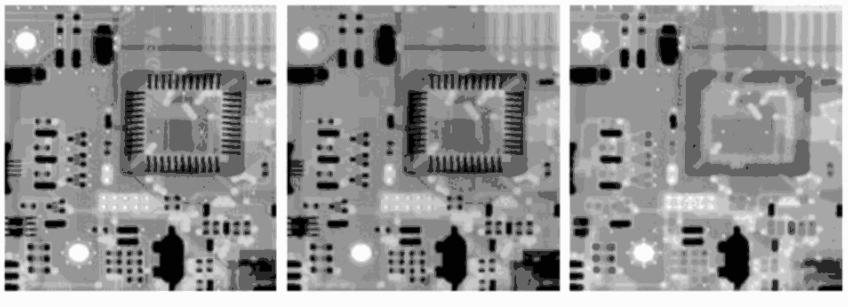
contrast stretching local

erosion karanlık alanları büyütüyor



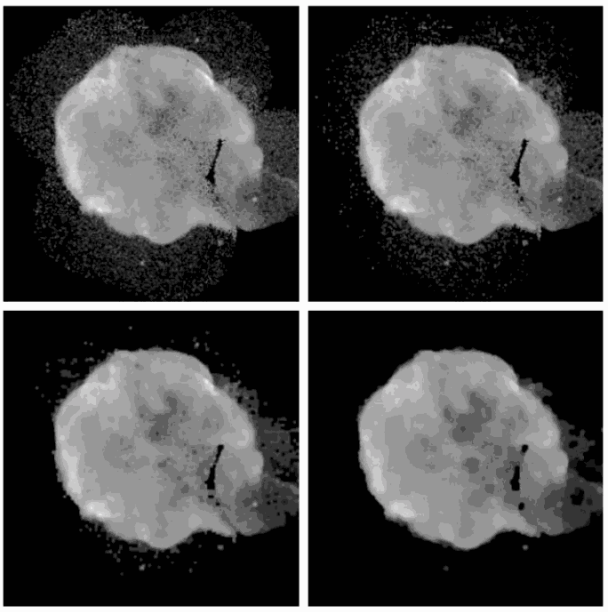
a b c

FIGURE 9.35 (a) A gray-scale X-ray image of size 448 × 425 pixels. (b) Erosion using a flat disk SE with a radius of two pixels. (c) Dilation using the same SE. (Original image courtesy of Lixi, Inc.)



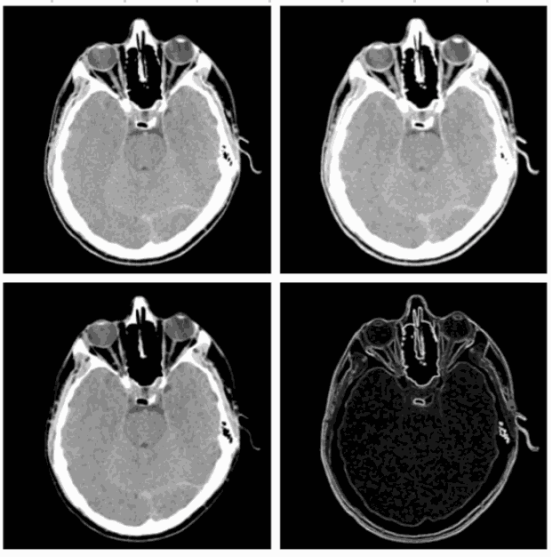
a b c

FIGURE 9.37 (a) A gray-scale X-ray image of size 448 × 425 pixels. (b) Opening using a disk SE with a radius of 3 pixels. (c) Closing using an SE of radius 5.



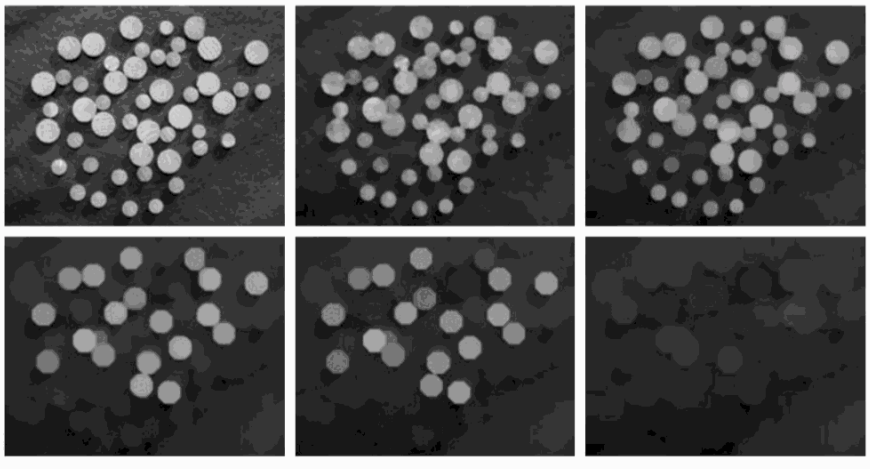
a b
c d

FIGURE 9.38
(a) 566 × 566 image of the Cygnus Loop supernova, taken in the X-ray band by NASA's Hubble Telescope. (b)–(d) Results of performing opening and closing sequences on the original image with disk structuring elements of radii, 1, 3, and 5, respectively. (Original image courtesy of NASA.)



a b
c d

FIGURE 9.39
(a) 512 × 512 image of a head CT scan. (b) Dilation. (c) Erosion. (d) Morphological gradient, computed as the difference between (b) and (c). (Original image courtesy of Dr. David R. Pickens, Vanderbilt University.)



a b c
d e f

FIGURE 9.41 (a) 531 × 675 image of wood dowels. (b) Smoothed image. (c)–(f) Openings of (b) with disks of radii equal to 10, 20, 25, and 30 pixels, respectively. (Original image courtesy of Dr. Steve Eddins, The MathWorks, Inc.)

Parçacik sayma Granulometry